

MODULE 33

React Component Architecture

Build reusable, scalable, and maintainable user interfaces with React components, JSX, props, and composition.







MODULE 33 OVERVIEW

Modern enterprise applications need fast, interactive, maintainable user interfaces -- React's component model is the industry-standard way to build them.

Enterprise systems like the Customer Management Platform CRM need a frontend that's as disciplined as the backend: typed, composable, accessible, and built from small, tested, reusable pieces.

DURATION & LAB



Module Duration

1 Hour Theory · 2.5 Hours Lab · 3.5 Hours Total



Theory Focus

React fundamentals, JSX, props, composition, and modular UI design.



Business Scenario

Build the Customer Management Platform CRM dashboard component shell.

WHY A COMPONENT-BASED FRONTEND MATTERS

Without Component-Based UI	With React Components
Monolithic, tangled UI code -- hard to maintain or extend.	Modular, composable UI -- small, reusable components.
Duplicated markup and logic copy-pasted across pages.	Write once, reuse everywhere via props and composition.
Manual DOM updates -- slow, error-prone, hard to reason about.	Declarative rendering -- the Virtual DOM updates only what changed.
Inconsistent UI and accessibility across teams.	Consistent, accessible, well-tested components (StatusBadge, CustomerCard).
Slow onboarding -- new developers learn ad-hoc conventions.	Faster onboarding -- clear component contracts and typed props.

KEY TAKEAWAY React's component architecture turns a tangled frontend into a modular, typed, accessible system built from small, reusable, well-tested pieces -- exactly what Lab 33's CRM dashboard requires.



WHY REACT IS POPULAR

React has become the go-to library for building user interfaces -- here's why developers and enterprises love it.



High Performance

Uses a Virtual DOM to update only what changed, keeping apps fast.



Component Based

Build reusable UI components and compose complex interfaces easily.



Declarative

Makes your code predictable and easier to debug and maintain.



Large Ecosystem

Huge community support and a rich ecosystem of libraries and tools.



Cross-Platform

React Native builds iOS and Android apps from shared logic.



Scalable for Any Size

Scales from small projects to large enterprise systems with ease.



Backed by Meta

Maintained by Meta, with continuous improvements and stability.



Developer Friendly

Easy to learn, with great docs and powerful DevTools.

INDUSTRY ADOPTION

Facebook

Instagram

Netflix

Airbnb

Tesla

Thousands of enterprises

KEY TAKEAWAY React empowers developers to build modern, interactive, and maintainable user interfaces faster -- while delivering exceptional user experiences.

MODULE LEARNING OBJECTIVES

By the end of this module, you will be able to develop well-structured, reusable, and dynamic React components.

- **Understand React Fundamentals** — explain what React is, its architecture, and the basic structure of a React application.
- **Build Components** — create functional components and understand the component lifecycle at a high level.
- **Work with JSX** — use JSX syntax to describe UI and render dynamic content in a React application.
- **Use Props Effectively** — pass data between components using props and make components reusable.
- **Compose Components** — combine components to build complex, modular, and maintainable user interfaces.
- **Design Modular UIs** — apply best practices to design scalable, maintainable, and enterprise-ready UIs.
- **Apply Best Practices** — follow React best practices for component structure, naming, and code organization.
- **Build in the Lab** — develop, test, and enhance a React application by building reusable components.

KEY TAKEAWAY You will be able to develop well-structured, reusable, and dynamic React components to build modern, user-friendly, and enterprise-grade web applications.

WHAT IS REACT?

React is a JavaScript library for building user interfaces, especially single-page applications where you need a fast, interactive experience.

KEY HIGHLIGHTS

- **Component Based** — build encapsulated, reusable UI components.
- **Declarative** — describe how the UI should look for a given state; React handles the updates.
- **Efficient** — uses a Virtual DOM to update only the parts of the UI that actually changed.
- **Flexible & Scalable** — an ecosystem of libraries and tools, from small widgets to large enterprise applications.

A SIMPLE EXAMPLE — Welcome.jsx

```
import React from "react";

function Welcome(props) {
  return <h1>Hello, {props.name}! 🙌</h1>;
}

export default Welcome;
```

React helps developers build reusable UI components and efficiently update and render the right components when data changes.

HOW REACT WORKS



Reusable Components

High Performance

Scalable Ecosystem

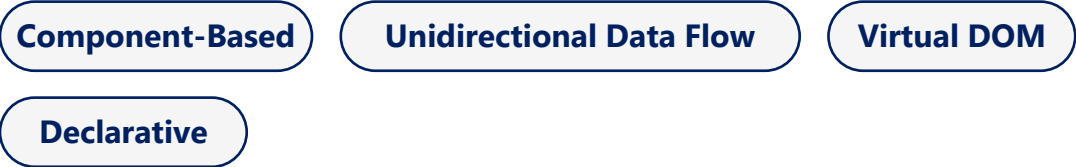
Trusted by Industry Leaders

KEY TAKEAWAY React is a powerful library to build fast, maintainable, and scalable user interfaces using reusable components and a highly efficient rendering model.

REACT ARCHITECTURE OVERVIEW

React follows a component-based architecture with a unidirectional data flow to build fast, scalable, maintainable user interfaces.

CORE PRINCIPLES



KEY BUILDING BLOCKS

Block	What It Does
Components	Building blocks of a React app; functional or class.
JSX	A syntax extension for writing HTML-like code in JS.
Props	Read-only inputs passed from parent to child.
State	Local data managed within a component, can change.
Hooks	Functions that let you use state and other features.
Context	Shares data across the component tree without prop drilling.

HOW DATA FLOWS IN REACT



THE THREE ARCHITECTURE LAYERS

- **UI Layer** — your components (App, Header, CustomerList, ...).
- **React Core** — lifecycle, reconciliation (diffing), the Virtual DOM.
- **Data Layer** — state, data, and APIs that feed the UI layer.

Data flows down via props; user events flow up via callbacks -- this predictable, one-directional loop is what makes large React apps easy to reason about.

KEY TAKEAWAY React is built around components and unidirectional data flow: props flow down, events flow up via callbacks -- an architecture that enables scalable, maintainable, high-performance apps.

REACT APPLICATION STRUCTURE

A well-structured React application makes your code easier to understand, maintain, and scale.

WHAT EACH FOLDER/FILE DOES

Folder/File	Purpose
public/	Static files served as-is; index.html is the main HTML file.
src/	Contains all the source code of the application.
assets/	Images, icons, fonts, and other static assets.
components/	Reusable UI components used across pages (Header, Button).
pages/	Page-level components for different routes/views.
App.jsx	The root component that composes the UI and defines routes.
index.jsx	Entry point; renders <App /> into the DOM.
package.json	Project metadata, dependencies, and scripts.

APP.JSX (EXAMPLE)

```
import Header from "../components/Header";
import Footer from "../components/Footer";
import Home from "../pages/Home";
function App() {
  return (
    <>
      <Header />
      <Home />
      <Footer />
    </>
  );
}
export default App;
```

BEST PRACTICES

- Keep components small
- Group files by feature
- Meaningful names
- Reuse components
- Consistent structure

KEY TAKEAWAY Components are organized by feature and reusability; index bootstraps the app; App.jsx composes the UI -- a structure that stays scalable and easy to maintain as the app grows.

FUNCTIONAL COMPONENTS

Functional components are JavaScript functions that return JSX -- simpler, easier to read, and the preferred way to write components in modern React.

KEY CHARACTERISTICS

- **JavaScript Function** — created using a plain JavaScript function.
- **Returns JSX** — the function returns JSX that describes what to render.
- **No this Keyword** — they don't have their own instance or this context.
- **Reusable** — accept props as input and can be reused anywhere.
- **Lightweight** — easier to test, maintain, and reason about.

COMPARISON: CLASS VS FUNCTIONAL

Aspect	Class	Functional
Syntax	ES6 class	Plain function
State	this.state	useState Hook
Lifecycle	Many methods	useEffect Hook
Code Size	More verbose	More concise
Recommended	Legacy/complex	Preferred in modern React

EXAMPLE: FUNCTIONAL COMPONENT — Welcome.jsx

```
import React from "react";

// Functional component
function Welcome(props) {
  return (
    <div className="welcome">
      <h2>Hello, {props.name}! 🙌</h2>
      <p>Welcome to React components.</p>
    </div>
  );
}
export default Welcome;
```

WHY USE FUNCTIONAL COMPONENTS?

Cleaner syntax

Easier to test

Work with Hooks

Better performance

Community recommended

KEY TAKEAWAY Functional components with Hooks are the modern, recommended way to write React -- cleaner syntax, easier testing, and full feature parity with class components.

TRY IT YOURSELF — EXERCISE 1: FILL COMPONENT TODOS (STARTER)

Reinforces: basic functional components, typed props, and JSX — fill in every blank before moving on.

FILL IN EVERY ____ (notes/lab33-todos.md)

CustomerCard.tsx (STARTER — not a solution)

```
type CustomerCardProps = {
  customerId: string;
  name: string;
  status: ____;
  onSelect: (id: string) => void;
};

export function CustomerCard(
  { customerId, name, status, onSelect }: CustomerCardProps
) {
  return (
    <article aria-label={____}>
      <h3>{____}</h3>
      <StatusBadge status={status} />
      <button type="button" onClick={() => onSelect(____)}>
        View
      </button>
    </article>
  );
}
```

#	What To Do
1	Paste the stub above into notes/lab33-todos.md exactly as shown.
2	Fill blanks with: 'ACTIVE' 'SUSPENDED', `\${name} (\${customerId})`, name, customerId.
3	Add a filled sample usage: <CustomerCard customerId="CUS-1001" name="Amina Khan" status="ACTIVE" onSelect={...} />.
4	Write an a11y note: the button has visible text; avoid icon-only buttons without aria-label.

This starter is intentionally simpler than Lab 33's real CustomerCard (no onEdit, no mailto link) -- master typed props and JSX here first.

KEY TAKEAWAY TRY IT NOW — This is a STARTER, not a solution: fill every blank in exercises/exercise-01-fill-react-component-todos.md before moving on.

COMPONENT LIFECYCLE OVERVIEW

The component lifecycle is the series of phases a component goes through, from being created to being removed from the DOM.

LIFECYCLE PHASES

- **1. Mounting** — the component is being created and inserted into the DOM.
- **2. Updating** — the component updates due to changes in props or state.
- **3. Unmounting** — the component is being removed from the DOM.

CLASS LIFECYCLE METHODS

Method	When It Runs
constructor()	Component initialized.
render()	Every time it renders.
componentDidMount()	After added to the DOM.
componentDidUpdate()	After props/state change.
componentWillUnmount()	Just before removal.

FUNCTIONAL EQUIVALENT: `useEffect`

```
import { useEffect } from "react";

function Example({ data }) {
  useEffect(() => {
    console.log("mounted");
    return () => {
      console.log("cleanup");
    };
  }, [data]); // re-runs when data changes

  return <div>Data: {data}</div>;
}
```

Understanding these phases helps manage resources, side effects, and performance -- and prevents memory leaks.

BEST PRACTICES

Side effects in `useEffect`

Clean up in the return function

Avoid unnecessary updates

Use Hooks for lifecycle-like behavior

KEY TAKEAWAY Functional components use the `useEffect` Hook to handle mounting, updating, and cleanup -- keep side effects in `useEffect` and clean up resources in its return function.



COMPONENT COMPOSITION

Component composition is the process of combining simple, reusable components to build complex user interfaces.

WHY COMPONENT COMPOSITION?

- **Reusability** — build once, use everywhere.
- **Maintainability** — smaller components are easier to maintain and update.
- **Scalability** — easily combine and extend components as your app grows.
- **Team Collaboration** — small, focused components are easier to develop in parallel.

IN THE CRM DASHBOARD — App.tsx

```
import { CustomerList } from "../components/CustomerList";

function App() {
  return (
    <AppLayout>
      <CustomerToolbar onAdd={handleAdd} />
      <CustomerList
        customers={seedCustomers}
        onEdit={handleEdit}
      />
    </AppLayout>
  );
}
```

KEY TAKEAWAY Compose small, focused components together to build powerful, flexible, maintainable UIs -- exactly the CustomerList → CustomerCard → StatusBadge composition you'll build in the lab.

COMPONENT HIERARCHY



In the lab, this maps directly to App → CustomerList → CustomerCard → StatusBadge -- each level composes the level below it, and data (props) flows straight down that same chain.

BEST PRACTICES

Single responsibility

Compose via props & children

Composition over inheritance

Favor reusability

TRY IT YOURSELF — EXERCISE 2: COMPONENT INVENTORY

Reinforces: breaking a UI into components with single responsibilities — an analysis exercise.

EXAMPLE STARTING POINT (≥ 5 COMPONENTS)

Component	Responsibility (≤ 6 words)
App	Root shell; composes the dashboard
CustomerList	Renders one card per customer
CustomerCard	Displays a single customer's info
StatusBadge	Shows a customer's status text
PageHeader	Displays page title and actions

STEPS (IN notes/lab33-components.md)

#	What To Do
1	Screen: imagine a Customer list showing Amina and Ravi with status badges.
2	Inventory: list at least 5 components (use the starting point at left, or your own).
3	One Responsibility: for each component, write a responsibility in 6 words or fewer.
4	Notes: save the finished inventory as notes/lab33-components.md.

Compare your inventory to Lab 33's real suggested files -- StatusBadge, CustomerCard, CustomerList, CustomerForm, AppLayout, CustomerToolbar -- most of what you listed already has a real counterpart there.

KEY TAKEAWAY TRY IT NOW — Complete Exercise 2 before continuing: exercises/exercise-02-component-inventory.md (workspace: examples/module-33-exercises).

REUSABLE COMPONENTS

Reusable components are generic, self-contained UI pieces that can be used in multiple places across an application.

GOOD REUSABLE COMPONENT CHARACTERISTICS

- **Single Responsibility** — does one thing well.
- **Configurable via Props** — accepts data and behavior through props.
- **Flexible** — works in different contexts and layouts.
- **Independent** — does not rely on external state or side effects.
- **Well Documented** — easy to understand and use.

EXAMPLE: BUTTON.JSX

```
function Button({ label, type = "primary", onClick }) {  
  const cls = type === "primary" ? "btn-primary" : "btn-secondary";  
  return (  
    <button className={`btn ${cls}`} onClick={onClick}>  
      {label}  
    </button>  
  );  
}  
export default Button;
```

HOW TO BUILD REUSABLE COMPONENTS



BEST PRACTICES

Name clearly

Sensible default props

Consistent styling

Test in isolation

Organize in a dedicated folder

KEY TAKEAWAY Reusable components help you build consistent, maintainable, and scalable React applications by maximizing code reuse and minimizing duplication.

WHAT IS JSX?

JSX (JavaScript XML) is a syntax extension for JavaScript that allows you to write HTML-like code inside JavaScript.

WHY JSX?

Readable

Expressive

Developer Friendly

Powerful

KEY POINTS

- JSX looks like HTML, but it is JavaScript.
- It allows embedding expressions using { }.
- Browsers do not understand JSX directly.
- Babel (or a similar tool) transpiles JSX into JavaScript before execution.
- React uses that output to create and update the DOM efficiently.

JSX is not a requirement of React, but it is the most common, and most readable, approach.

JSX EXAMPLE

```
const name = "Amina";
const element = (
  <div className="card">
    <h1>Hello, {name}!</h1>
  </div>
);
```

TRANSPILED TO JAVASCRIPT

```
const element = React.createElement(
  "div", { className: "card" },
  React.createElement("h1", null, "Hello, ", name, "!")
);
```

JSX (you write)

Babel (transpiles)

JavaScript (browser runs)

KEY TAKEAWAY JSX brings HTML-like syntax into JavaScript, making React components more intuitive, readable, and maintainable. It enhances productivity and accelerates UI development.

JSX SYNTAX RULES (1 OF 2)

JSX follows specific syntax rules that ensure your code is valid and can be transpiled to regular JavaScript.

Rule	Correct	Incorrect
1. Return a single root element	<code>return (<div> <h1>Hi</h1> <p>Bye</p> </div>);</code>	<code>return (<h1>Hi</h1> <p>Bye</p>);</code>
2. Close all tags	<code></code>	<code></code>
3. Use className instead of class	<code><div className="container"></code>	<code><div class="container"></code>
4. Use htmlFor instead of for	<code><label htmlFor="name">Name</label></code>	<code><label for="name">Name</label></code>
5. Use curly braces for expressions	<code><h1>Hello, {userName}!</h1></code>	<code><h1>Hello, userName!</h1></code>
6. Use camelCase for attributes	<code><input readOnly tabIndex={0} /></code>	<code><input readonly tabIndex="0" /></code>

WHY THESE RULES MATTER

JSX compiles to JavaScript function calls -- a function can only return one value, tags map to real DOM elements that must be well-formed, and JSX attributes are JavaScript property names, which is why className/htmlFor/camelCase replace their HTML equivalents.

KEY TAKEAWAY Following JSX syntax rules ensures your components work correctly and your code stays clean, readable, and error-free.

JSX SYNTAX RULES (2 OF 2)

Four more essential rules, plus a quick-reference table of common JSX patterns.

ADDITIONAL JSX RULES

- **7. Quote string literals** — title="Hello" or title='Hello' -- never title=Hello.
- **8. Key for list items** — {items.map(i => <li key={i.id}>{i.name})} -- a unique, stable key on every mapped element.
- **9. Comments** — use {/* comment */} for multi-line comments inside JSX.
- **10. Render nothing with null** — return null; renders nothing at all -- not an empty string, not undefined.

COMMON EXAMPLES

Purpose	JSX Syntax
Image	
Link	React Docs
Button	<button onClick={handleClick}>Click</button>
Input	<input value={name} onChange={...} />
Conditional	{isLoggedIn ? <Dashboard /> : <Login />}
List Rendering	{items.map(i => <Item key={i.id} {...i} />)}

KEY TAKEAWAY Following JSX syntax rules ensures your components work correctly and your code remains clean and readable -- especially the key rule, which the Component Reusability and Lab 33 slides return to.

TRY IT YOURSELF — EXERCISE 3: JSX ON PAPER

Reinforces: sketching a JSX tree by hand with correct keys — a documentation exercise, no runtime needed.

TARGET SHAPE — SKETCH THIS ON PAPER FIRST

CustomerList (hand-written JSX tree)

```
<CustomerList>
  <CustomerCard key="CUS-1001">
    <h3>Amina Khan</h3>
    <StatusBadge status="ACTIVE" />
  </CustomerCard>
  <CustomerCard key="CUS-1002">
    <h3>Ravi Singh</h3>
    <StatusBadge status="PROSPECT" />
  </CustomerCard>
</CustomerList>
```

STEPS (IN notes/lab33-jsx-tree.md)

#	What To Do
1	Tree: sketch <CustomerList> containing two <CustomerCard> nodes, by hand.
2	Keys: write why key={customerId} should be "CUS-1001", not the array index.
3	Badge: nest <StatusBadge status="ACTIVE" /> inside Amina's card.
4	No Runtime: do not create a Vite app for this exercise -- paper only.

This exact tree — two <CustomerCard> nodes keyed by customerId, each nesting a <StatusBadge> — is what Lab 33's CustomerList.tsx renders from the seed fixtures array.

KEY TAKEAWAY TRY IT NOW — Complete Exercise 3 before continuing: exercises/exercise-03-jsx-paper.md (workspace: examples/module-33-exercises).

RENDERING DYNAMIC CONTENT

React makes it easy to render dynamic data in your UI by embedding JavaScript expressions inside JSX using curly braces {}.

EXAMPLES OF RENDERING DYNAMIC CONTENT

Kind	JSX Pattern
Variable	<code>const name = "Amina"; <p>Hello, {name}!</p></code>
Expression	<code>const a=10,b=20; <p>Sum is {a + b}</p></code>
Object Property	<code><p>{user.name} is a {user.role}</p></code>
Array (List)	<code>{items.map(i => <li key={i}>{i})}</code>
Conditional	<code>{isActive ? "Active" : "Inactive"}</code>

WAYS TO RENDER DYNAMIC CONTENT

- Variables
- Expressions
- Object Properties
- Arrays & Lists
- Conditional Rendering

COMPLETE EXAMPLE — CustomerCard.tsx

```
function CustomerCard({ customer }: Props) {
  return (
    <article>
      <h3>{customer.fullName}</h3>
      <p>Status: {customer.status}</p>
      <ul>
        {customer.tags.map((tag) => (
          <li key={tag}>{tag}</li>
        ))}
      </ul>
    </article>
  );
}
```

Keep logic in JavaScript, JSX for presentation; use meaningful keys; handle null/undefined data safely.

KEY TAKEAWAY By embedding expressions inside JSX, React lets you render dynamic data efficiently and keeps your UI in sync with application state and props.

UNDERSTANDING PROPS

Props (short for "properties") are read-only inputs passed from parent to child components to make components dynamic and reusable.

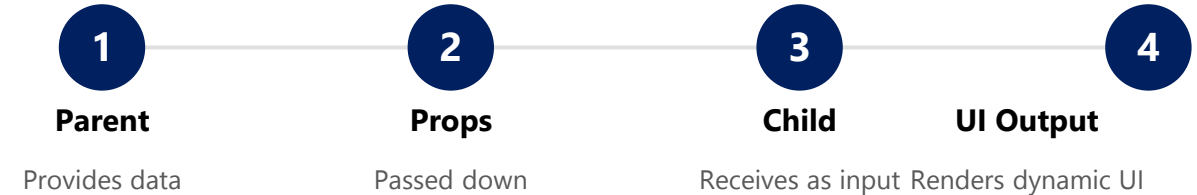
KEY POINTS

- Props flow in one direction: Parent → Child.
- Props are immutable (read-only) inside the child -- a child can never modify its own props.
- Props can carry any data: string, number, boolean, array, object, even functions.
- Access props via the props object, or (recommended) via destructuring.

ACCESSING PROPS — CustomerCard.tsx

```
interface Props {  
  customer: Customer;  
  onEdit: (id: string) => void;  
}  
// Destructuring (recommended)  
function CustomerCard({ customer, onEdit }: Props) { ... }
```

HOW PROPS WORK



EXAMPLE: PASSING PROPS — App.tsx

```
<CustomerCard  
  customer={{ customerId: "CUS-1001", fullName: "Amina Khan",  
    status: "ACTIVE" }}  
  onEdit={handleEdit}  
>
```

Read-only

Any data type

Parent → Child

Destructuring recommended

KEY TAKEAWAY Props are the primary way to pass data from parent to child components -- they are read-only and enable reusable, dynamic, and maintainable components.

TRY IT YOURSELF — EXERCISE 4: PROPS SKETCH

Reinforces: defining a typed props contract for CustomerCard — an architecture exercise.

CUSTOMERCARD PROPS REFERENCE

Prop	Amina Example	Ravi Example
customerId	CUS-1001	CUS-1002
name	Amina Khan	Ravi Singh
status	ACTIVE	PROSPECT
onSelect	() => void	() => void

STEPS (IN notes/lab33-props.md)

#	What To Do
1	Copy Props: copy the table at left; you've already added the Ravi (CUS-1002) row.
2	Types: write TypeScript-ish types, e.g. status: 'ACTIVE' 'SUSPENDED' 'PROSPECT'.
3	Children?: decide, in one sentence, whether CustomerCard takes children or props only.
4	Anti-Pattern: note -- never pass the entire global store as one mega-prop.

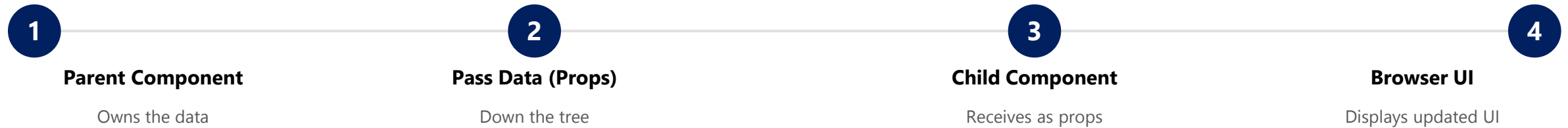
Lab 33's real CustomerCard uses exactly this shape: `interface Props { customer: Customer; onEdit: (id: string) => void; }` -- your sketch previews that same typed contract.

KEY TAKEAWAY TRY IT NOW — Complete Exercise 4 before continuing: `exercises/exercise-04-props-sketch.md` (workspace: `examples/module-33-exercises`).

PASSING DATA BETWEEN COMPONENTS (1 OF 2)

React uses one-way data flow to keep your application predictable and easy to manage. Data flows down from parent to child using props.

DATA FLOW PRINCIPLE



1. PASSING DATA DOWN (PARENT TO CHILD)

The parent sends data to the child component using props.

```
// Parent
<CustomerCard customer={amina} onEdit={handleEdit} />

// Child
function CustomerCard({ customer }: Props) {
  return <h3>{customer.fullName}</h3>;
}
```

WHY THIS APPROACH?

- **Makes components reusable** — the same child renders differently for whatever props it's given.
- **Easy to debug and test** — trace a bug by following props down one direction only.
- **Predictable data flow** — you always know where data originates.
- **Better performance** — React can optimize re-renders when data flow is one-directional.

KEY TAKEAWAY Use props to pass data from parent to child, in one predictable direction -- this keeps your app easy to debug, test, and scale.

PASSING DATA BETWEEN COMPONENTS (2 OF 2)

To send data back up, children use callbacks passed down as props. Siblings that share data lift state up to a common parent.

2. PASSING DATA UP (CHILD TO PARENT)

The child sends data back to the parent by calling a callback function the parent passed down as a prop.

```
// Parent owns the data and the handler
function App() {
  const handleEdit = (id: string) =>
    console.log("edit", id, "lab-request-001");
  return <CustomerCard customer={amina} onEdit={handleEdit} />;
}
// Child calls the callback prop -- never mutates data itself
<button onClick={() => onEdit(customer.customerId)}>Edit</button>
```

3. SIBLING COMMUNICATION

Siblings cannot talk directly. Lift the shared data up to their common parent, which then passes it down to both as props.

WHY THIS APPROACH?

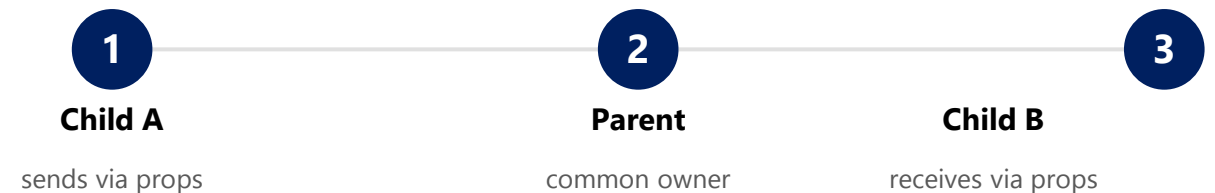
Makes components reusable

Easy to debug and test

Predictable data flow

Better performance

LIFTING STATE UP



This is exactly what Lab 34 will do to this lab's presentational components: lift the customer array into `useState()` in `App`, without rewriting any `CustomerCard` markup -- because the props CONTRACT (customer, `onEdit`) stays stable.

KEY TAKEAWAY Use props to pass data down; use callbacks to pass data up; lift state up when siblings need to share data -- one-way data flow keeps your app predictable, maintainable, and easy to scale.

COMPONENT REUSABILITY

Reusability is a core advantage of React. You can build a component once and use it many times with different data and behavior.

WHY REUSE COMPONENTS?

- **Avoid Duplication** — write once, use everywhere.
- **Save Time** — build features faster with existing building blocks.
- **Easy Maintenance** — update in one place, reflected everywhere it's used.
- **Consistency** — the same look and feel across the whole application.
- **Scalability** — easier to manage as the application grows.

EXAMPLES OF REUSABLE COMPONENTS

Category	Examples
UI Components	Button, Card, Input, Modal, StatusBadge
Layout Components	Header, Sidebar, Footer, AppLayout
Form Components	TextInput, Select, Checkbox, CustomerForm
Data Components	Table, List, Pagination, CustomerList
Feedback Components	Alert, Toast, Spinner, EmptyState

TIPS FOR BUILDING REUSABLE COMPONENTS

Small & focused

Customize via props

Sensible defaults

Composable

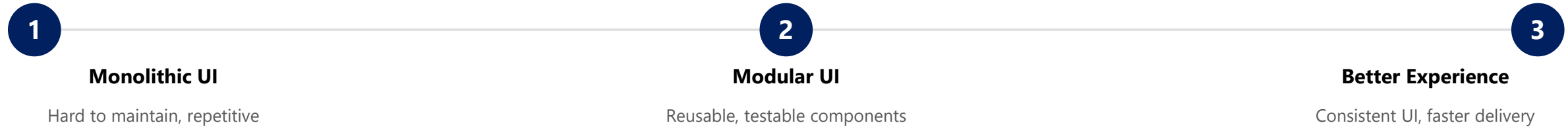
Avoid hardcoding

KEY TAKEAWAY Reusable components make your React applications more efficient, consistent, and maintainable. Build once, use everywhere.

DESIGNING MODULAR USER INTERFACES (1 OF 2)

Modular UI design means breaking a user interface into small, reusable components that work together to build complex, maintainable applications.

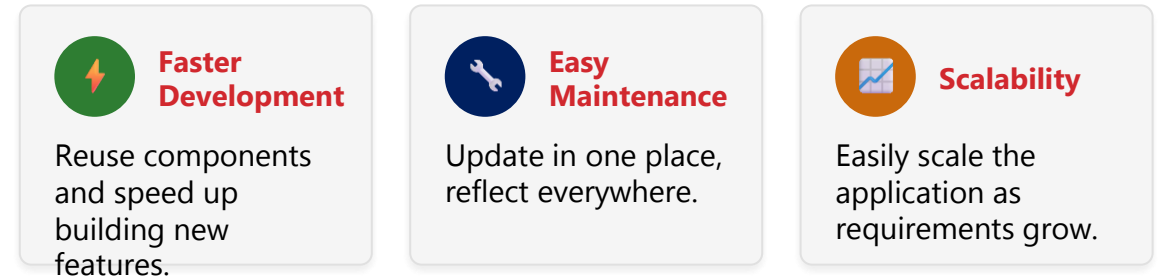
FROM MONOLITH TO MODULAR



PRINCIPLES OF MODULAR UI DESIGN

- **Single Responsibility** — each component should do one thing and do it well.
- **Reusability** — build components that can be used in multiple places.
- **Composition Over Inheritance** — combine small components to build complex UIs.
- **Consistent Design** — follow a design system for colors, typography, spacing.
- **Separation of Concerns** — keep UI, logic, and data separated.

BENEFITS



DESIGN SYSTEM INTEGRATION



KEY TAKEAWAY Designing modular user interfaces with reusable components makes your React applications cleaner, easier to maintain, and delightful for users.

TRY IT YOURSELF — EXERCISE 5: A11Y CHECKLIST

Reinforces: planning accessibility checks before coding the CRM dashboard — a documentation exercise.

ACCESSIBILITY CHECKS TO PLAN (notes/lab33-a11y-checklist.md)

Check	What To Confirm
1 — Semantics	Prefer <button>, <h1>-<h3>, and / over clickable <div>s with onClick handlers.
2 — Contrast	Status colors need visible text or an icon/shape alongside them, never color alone.
3 — Keyboard	Confirm Tab order can reach the View action for both Amina's and Ravi's cards.
4 — Checklist File	Save five checkbox lines covering these checks in notes/lab33-a11y-checklist.md.

WHAT GOOD A11Y LOOKS LIKE IN JSX

CustomerCard.tsx (accessible shape)

```
<article aria-label={` ${name} (${customerId})`} >
  <h3 id={`customer-${customerId}`} >{name}</h3>
  <StatusBadge status={status} />  {/* renders TEXT, not color alone */}
  <button type="button" onClick={() => onSelect(customerId)}>
    View
  </button>
</article>
```

KEY TAKEAWAY TRY IT NOW — Complete Exercise 5 before continuing: exercises/exercise-05-a11y-checklist.md (workspace: examples/module-33-exercises).

DESIGNING MODULAR USER INTERFACES (2 OF 2)

A worked example: breaking a dashboard UI into a component hierarchy, then composing it back together in code.

1. BREAK DOWN THE UI INTO A HIERARCHY

- App
 - AppLayout (Header, main landmark)
 - CustomerToolbar (Add button)
 - CustomerList
 - CustomerCard (repeated, keyed by customerId)
 - StatusBadge
 - CustomerForm

2. COMPONENT COMMUNICATION

- **Parent → Child** — App passes customers down to CustomerList via props.
- **Child → Parent** — CustomerCard sends edit intent up via the onEdit callback.

3. EXAMPLE CODE (COMPOSITION) — App.tsx

```
function App() {  
  return (  
    <AppLayout>  
      <CustomerToolbar onAdd={handleAdd} />  
      <CustomerList  
        customers={seedCustomers}  
        onEdit={handleEdit}  
      />  
      <CustomerForm value={draft} errors={{}}  
        onChange={handleChange} onSubmit={handleSubmit}  
        onCancel={handleCancel} />  
    </AppLayout>  
  );  
}
```

BEST PRACTICES

Start small

Meaningful names

Extract repeated UI

Document components

KEY TAKEAWAY Break the UI into components, build each reusable piece, compose them together, and deliver a great, consistent user experience -- this is exactly Lab 33's build order.

REACT BEST PRACTICES (1 OF 2)

Following best practices helps you build React applications that are maintainable, scalable, performant, and easy to work with.

#	Practice	What It Means
1	Use Functional Components	Prefer functional components with Hooks over class components.
2	Use Hooks Effectively	Use state and effect Hooks appropriately; avoid unnecessary effects.
3	Keep Components Small and Focused	Each component should do one thing well (Single Responsibility).
4	Use Props for Communication	Pass data via props; keep components pure and predictable.
5	Optimize Performance	Use React.memo, useMemo, useCallback to prevent unnecessary re-renders.
6	Organize Your Code	Follow a clear folder structure and consistent naming conventions.
7	Handle Errors Gracefully	Use error boundaries and proper fallback UI.
8	Write Clean and Consistent Code	Use ESLint, Prettier, and a consistent code style.

CORE PRINCIPLES TO FOLLOW

Keep It Simple

Reusability

Performance

Reliability

Consistency

KEY TAKEAWAY Good practices today lead to better code, faster development, and fewer bugs tomorrow -- easier to maintain, better performance, fewer bugs and issues.

REACT BEST PRACTICES (2 OF 2)

Best practices in code: a functional component with a Hook, a reusable component with props, and a recommended folder structure.

FUNCTIONAL + A HOOK — Counter.jsx

```
import { useState } from "react";
function Counter() {
  const [count, setCount] = useState(0);
  return (
    <button onClick={() => setCount(count + 1)}>
      Count: {count}
    </button>
  );
}
```

REUSABLE WITH PROPS — Button.jsx

```
function Button({ label, variant = "primary", onClick }) {
  return (
    <button className={`btn btn-${variant}`} onClick={onClick}>
      {label}
    </button>
  );
}
// <Button label="Save" variant="primary" onClick={save} />
```

RECOMMENDED COMPONENT STRUCTURE

- **components/** — reusable UI components (Button, StatusBadge, CustomerCard).
- **pages/** — page-level components (Dashboard, Profile).
- **hooks/** — custom React Hooks (useFetch).
- **utils/** — utility functions and API calls.
- **App.jsx / index.jsx** — root component and entry file.

ADDITIONAL TIPS

Meaningful names

Avoid deep nesting

Controlled forms

Keys on lists

Test components

KEY TAKEAWAY Following best practices helps you build React applications that are clean, efficient, scalable, and delightful for users and developers alike.

TRY IT YOURSELF — EXERCISE 6: LAB 33 READINESS CHECKLIST

Capstone pre-lab check — confirm component contracts and workspace before opening Lab 33.

READINESS CHECKLIST (RECORD IN `notes/lab33-readiness.md`)

Check	What To Confirm
Folder Plan	Note the future path: <code>crm-ui/src/components/</code> (created in the lab, not now).
Evidence	Plan a future screenshot: the list view showing Amina + Ravi cards (lab, not now).
Boundary	Confirm no <code>npm install</code> or <code>npm start</code> is required for this pre-lab.
Pass Mark	Pass if your Exercise 4 (props) and Exercise 1 (TODOs) notes both exist.

Setup (PowerShell) — from `EXERCISES-INDEX.md`

```
cd $env:USERPROFILE\java-bootcamp
New-Item -ItemType Directory -Force -Path examples\module-33-exercises | Out-Null
cd examples\module-33-exercises
java -version
```

KEY TAKEAWAY TRY IT NOW — Complete Exercise 6 before opening Lab 33: `exercises/exercise-06-lab33-readiness.md`.

LAB OVERVIEW -- REACT COMPONENTS FOR THE CRM DASHBOARD

Lab 33: React Components for the CRM Dashboard -- the Customer Management Platform React client.

BUSINESS SCENARIO

- Before state (Lab 34) and API integration (Lab 35), leadership freezes the UI contract: every customer surface must be typed, composable, and accessible -- color-only status, index keys, and class-name-only tests are all rejected.
- You own the gate for Amina (ACTIVE) and Ravi (PROSPECT) cards, empty-list UX, and an accessible form shell.
- This lab builds the presentational shell only -- fixtures, no API yet -- so Lab 34 can lift state without rewriting markup.

LEARNING OBJECTIVES

- Scaffold a React TypeScript CRM app with Vite.
- Define typed Customer, CustomerStatus, and CustomerDraft models.
- Create accessible StatusBadge and CustomerCard components.
- Compose CustomerList from cards with stable customerId keys.
- Build a labeled CustomerForm and empty/loading/error shells.
- Write React Testing Library tests that query by role, not class name.

WHAT YOU BUILD

- Vite React-TS crm-ui under lab33-crm; typed models + seed fixtures Amina/Ravi; StatusBadge, CustomerCard, CustomerList, CustomerForm, layout shells; empty/loading/error presentation components; RTL tests; npm run build success.

KEY TAKEAWAY Fixtures CUS-1001 (Amina Khan, ACTIVE) and CUS-1002 (Ravi Singh, PROSPECT) anchor every example; correlation id lab-request-001 appears on every logged edit/add callback.

LAB TASKS AND DELIVERABLES

React Components for the CRM Dashboard -- implementation steps, deliverables, and the evaluation rubric.

#	Implementation Step
1	Scaffold Vite React-TS crm-ui; install Vitest + React Testing Library.
2	Define Customer / CustomerStatus / CustomerDraft types + seed fixtures Amina/Ravi.
3	Create StatusBadge -- visible text status, not color alone.
4	Create CustomerCard -- <article>, heading id, mailto link, Edit button.
5	Compose CustomerList with stable customerId keys + EmptyState.
6	Create labeled CustomerForm with htmlFor/id pairs + role="alert" errors.
7	Compose the dashboard: AppLayout, CustomerToolbar, one main landmark.
8	Add LoadingState / ErrorState shells; write RTL tests by role; evidence pack.

EXPECTED DELIVERABLES

- Complete crm-ui project, organized folder structure.
- Reusable components with props and clean code.
- Parent-to-child and child-to-parent data flow.
- RTL tests green; npm run build succeeds.
- README runbook + component notes + screenshots.

RUBRIC (100 MARKS)

- Environment 10 · Core Impl 30 · Integration/Config 15 · Failure Handling 15 · Verification 10 · Security/A11y 10 · Docs 10

KEY TAKEAWAY Class-name-only tests lose automated-verification marks; index keys are an honor violation; color-only status without visible text is an accessibility failure.

MODULE 33 SUMMARY

In this module, we learned how to build modular, reusable, and maintainable user interfaces in React by working with components, props, JSX, and best practices.

KEY CONCEPTS COVERED



JSX Syntax

Write HTML-like code in JavaScript using JSX, transpiled by Babel.



Functional Components

Build components as plain functions that return JSX.



Understanding Props

Pass typed, read-only data from parent to child components.



Passing Data

Data down via props, events up via callbacks -- one-way data flow.



Composition & Reuse

Combine small, reusable, well-documented components into complex UIs.



Best Practices

Modular design, accessibility, and clean, consistent code organization.

MODULE FLOW RECAP

1

JSX & Components

2

Props & Data Flow

3

Composition & Reuse

4

Modular UI Design

5

Lab: CRM Dashboard

KEY TAKEAWAY React components are the building blocks of modern user interfaces. By mastering JSX, props, composition, and best practices, you can build powerful, reusable, and scalable applications.

KNOWLEDGE CHECK (1 OF 3)

Test your understanding of React components, props, state, and JSX -- questions 1-4 of 10.

1. What is the primary purpose of a React component?

- A. To store data in the browser
- B. To define reusable pieces of UI**
- C. To handle routing in the application
- D. To connect to the database

3. How do you access a prop named "title" in a functional component?

- A. props.title**
- B. this.props.title
- C. title.props
- D. state.title

2. Which of the following is used to pass data from a parent to a child component?

- A. useState
- B. useEffect
- C. Props**
- D. useContext

4. Which Hook is used to manage local state in a functional component?

- A. useEffect
- B. useState**
- C. useReducer
- D. useContext

KEY TAKEAWAY A component defines a reusable piece of UI; props pass data down; access a prop via props.title (functional components have no this); useState manages local state.

KNOWLEDGE CHECK (2 OF 3)

Four more questions on re-renders, reusability, callbacks, and modular design -- questions 5-8 of 10.

5. What will trigger a re-render in a functional component?

- A. Updating state using useState
- B. Changing a prop value
- C. Calling a regular JavaScript function
- D. All of the above**

7. What is the correct way to pass a function from a parent to a child component?

- A. By using props**
- B. By using state
- C. By using context
- D. By using ref

6. Which of the following helps in making components reusable?

- A. Hardcoding values inside the component
- B. Using props to accept dynamic data**
- C. Duplicating the component code
- D. Using global variables

8. Which principle suggests breaking the UI into small, independent pieces?

- A. Separation of Concerns
- B. Component Reusability
- C. Modular UI Design
- D. All of the above**

KEY TAKEAWAY State changes, prop changes, and calling a regular function can all trigger a re-render; props (not global variables) make components reusable and pass callbacks down.

KNOWLEDGE CHECK (3 OF 3)

The final two questions, on maintainability and file conventions -- questions 9-10 of 10.

9. Which practice helps keep components easy to maintain?

- A. Writing large components with multiple responsibilities
- B. Using meaningful names and keeping components small**
- C. Avoiding props to reduce complexity
- D. Using inline styles everywhere

10. Which file extension is commonly used for React components?

- A. .js
- B. .jsx**
- C. .ts
- D. .html

KEY TAKEAWAY Meaningful names and small, focused components stay maintainable; .jsx (and .tsx for typed components, as in Lab 33) is the conventional React component file extension.

MODULE 33 COMPLETE

React Component Architecture

You can now build functional React components, write JSX, pass data with typed props, compose reusable pieces, and design modular, accessible user interfaces.